

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 3 – Mock Interview
Weightage:	30% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	M. Sneha	Reg. No.:	192425092
Section / Batch:	2024- AI&ML	Date:	

Code of Conduct

I M. Sneha(192425092)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M. Sneha_____

Instructions

- Start the mock interview by entering the provided ChatGPT prompt exactly as given in the Annexure A in the next page. Enter your Name and Register Number when requested.
- Answer all 10 interview questions independently and in your own words using mobile chatgpt voice. Do not copy content from external sources or use AI-generated responses during the interview.
- Provide detailed and meaningful answers with appropriate technical terminology, examples, and justifications wherever applicable. One-line answers may lead to lower scores.
- Complete the interview in a single session. Ensure that all 10 questions are answered before requesting the final evaluation and assessment report.
- Submit the final ChatGPT-generated report printout, including the interview questions, your responses, question-wise rubric evaluation, and the Student Average Final Mark, as proof of completion.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse & Reference Resolution	Understanding discourse, anaphora resolution, reference tracking, and contextual interpretation in text	Q1
Text Coherence & Discourse Structure	Analyzing coherence relations, discourse organization, and maintaining meaningful text flow	Q2
Dialog Acts & Interpretation	Identifying dialog acts, speaker intentions, and conversational meaning	Q3
Dialog and Conversational Agents	Architecture, components, and working of conversational agents/chatbots	Q4
Language Generation Architecture	Stages of Natural Language Generation (NLG), pipeline architecture, and applications	Q5
Surface Realization & Discourse Planning	Transforming semantic representations into grammatically correct sentences and planning discourse	Q6
Machine Translation – Transfer Approach	Translation process using transfer-based methods and practical applications	Q7
Machine Translation – Interlingua Approach	Language-independent representation and multilingual translation framework	Q8
Statistical Approaches in NLP	Statistical language models, probability-based language generation, and machine translation	Q9

Integrated Case Study (Analytical & Problem Solving)	Real-world scenario involving discourse analysis, conversational agents, language generation, and machine translation	Q10
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Annexure A – Mock Interview

MOCK INTERVIEW ASSESSMENT REPORT

Student Name: M. Madhu Sneha

Register Number: 192425092

INTERVIEW QUESTIONS AND RESPONSES Q1:

Question Asked: What is the transfer approach in machine translation?

Student Response: The transfer approach translates a source-language sentence into a target language using an intermediate structural representation. It handles grammatical and word-order differences between the two languages.

Q2:

Question Asked: Explain the three major stages of a transfer-based machine translation system.

Student Response: The three stages are **Analysis, Transfer, and Generation**. Analysis studies the source sentence, Transfer changes it to the target structure, and Generation produces the final sentence.

Q3:

Question Asked: Compare direct translation with transfer-based translation.

Student Response: Direct translation translates words or phrases with minimal analysis. Transfer-based translation performs deeper grammatical analysis and handles structural differences between languages.

Q4:

Question Asked: Explain how a transfer-based system could translate a sentence from English to Hindi while accounting for differences in word order.

Student Response: The system analyzes the English sentence and identifies its grammatical parts. During transfer, it changes the **SVO** structure into Hindi's **SOV** structure before generating the Hindi sentence.

Q5:

Question Asked: Why is linguistic transfer necessary when two languages have different grammatical structures?

Student Response: Linguistic transfer is needed because languages have different grammar, word order, and sentence structures. It helps convert the source structure into a grammatically correct target structure.

Q6:

Question Asked: An English sentence follows Subject–Verb–Object, while the target language follows Subject–Object–Verb. Explain how the transfer stage can transform the sentence structure before generation.

Student Response: The transfer stage identifies the subject, verb, and object and rearranges them from **SVO to SOV**. The generation stage then produces the correct target-language sentence.

Q7:

Question Asked: What is the Interlingua approach in Machine Translation?

Student Response: Interlingua is a language-independent intermediate representation used in machine translation. The source language is converted into Interlingua and then into the target language.

Q8:

Question Asked: Compare the transfer approach and Interlingua approach.

Student Response: Transfer uses language-pair-specific rules, while Interlingua uses one common language-independent representation. Interlingua is therefore more suitable for multilingual translation.

Q9:

Question Asked: Analyze one major advantage and one major limitation of using Interlingua for multilingual translation.

Student Response: The main advantage of Interlingua is that it simplifies multilingual translation by using a common representation. Its limitation is that designing a universal representation for all languages is complex.

Q10:

Question Asked: Suppose a company needs translation between English, Hindi, Tamil, French, and German. Explain how an Interlingua-based system can reduce the number of translation components required.

Student Response: For 5 languages, pairwise translation needs **20 directional components**, while Interlingua needs only **10 language-to/from-Interlingua components**. Thus, Interlingua reduces system complexity.

RUBRIC EVALUATION

Criterion	Weightage	Assigned Level	Marks Awarded
Technical Knowledge	25	Level X	XX
Problem Solving	25	Level X	XX
Analytical Thinking	20	Level X	XX
Communication	15	Level X	XX
Confidence	15	Level X	XX

SUMMARY TABLE

Technical Knowledge (25)	Problem Solving (25)	Analytical Thinking (20)	Communication (15)	Confidence (15)	SubQ Total	Student Avg Final Mark
Level X	Level X	Level X	Level X	Level X	XX.X	XX.X

OVERALL FEEDBACK

Strengths:

List strengths Areas for Improvement:

List improvements Interview Readiness:

Excellent / Good / Average / Needs Improvement End the report after the Interview Readiness section.

After evaluating each answer, do NOT display the marks immediately. Store the evaluation internally.

After Question 10 is completed:

Generate a Question-wise Evaluation Table for all 10 questions.

For each question assign:

Technical Knowledge Level

Problem Solving Level

Analytical Thinking Level

Communication Level

Confidence Level

SubQ Total

Calculate SubQ Total using: Level 4 = 100% weightage

Level 3 = 75% weightage

Level 2 = 50% weightage

Level 1 = 25% weightage

Calculate Student Avg Final Mark as:

Average of all 10 SubQ Totals

Display:

Complete interview questions

Student answers

Question-wise evaluation table

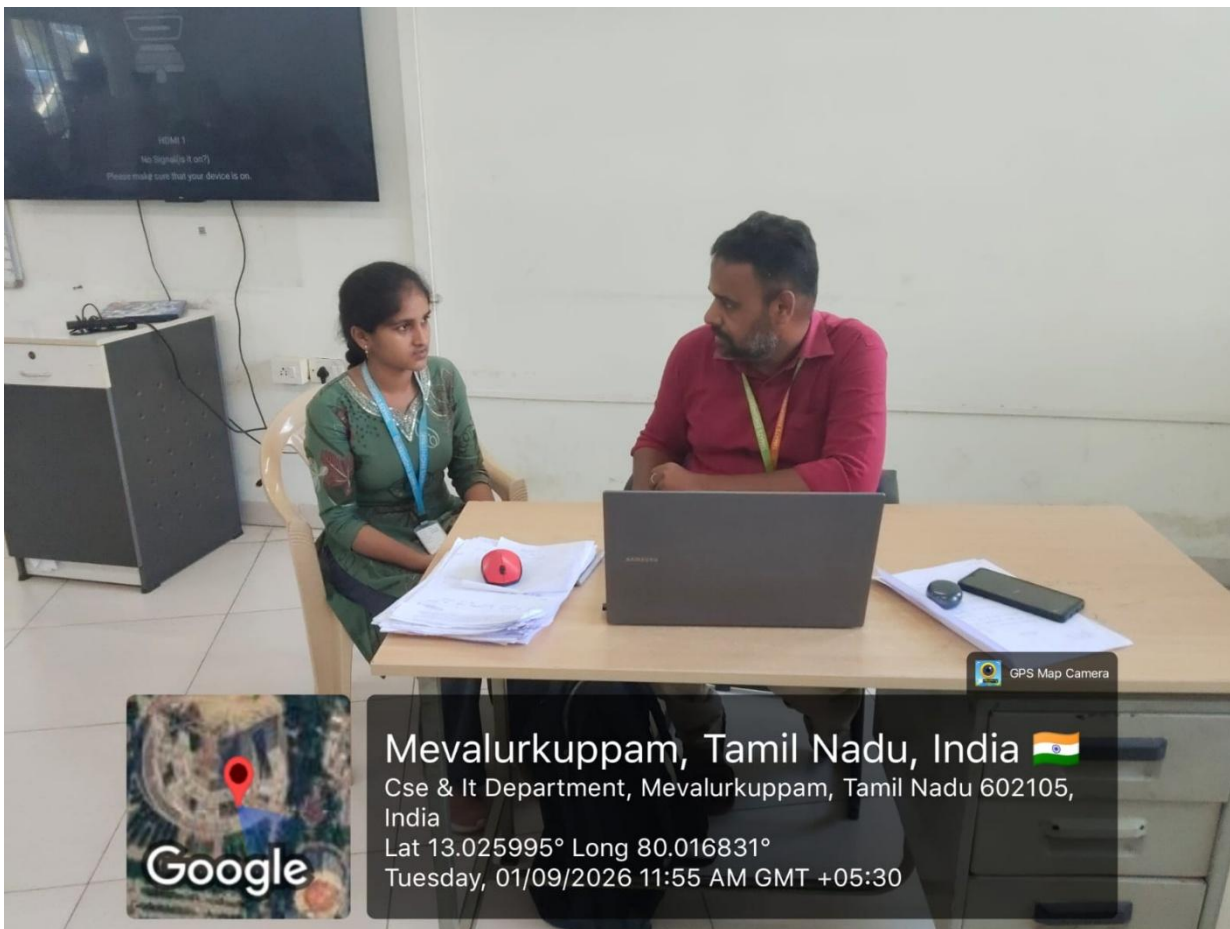
Student Avg Final Mark

Overall strengths

Areas for improvement

Interview readiness

Geotag Photo:



Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	20	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1	3	3	4	4	4	18	86
2	4	3	3	3	3	16	
3	4	4	4	3	3	18	
4	3	3	4	3	3	16	
5	4	3	4	3	4	18	
6	3	3	3	3	4	16	
7	4	3	3	4	4	18	
8	4	3	3	4	3	17	
9	3	4	3	4	4	18	
10	3	4	4	3	3	17	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: ____	Date: ____	Date: ____